

Sampling distributions, standard errors, and the mini-exam



Overview

Review:

- Sampling distributions and the standard error
- Sampling distributions in R

If there is time: confidence intervals

The mini-exam: 1:45-2:15 (5 minute break at 1:40)

Announcement

Homework 4 is due on Gradescope on **Sunday October 4th at 11pm**

- **Be sure to mark each question on Gradescope!**

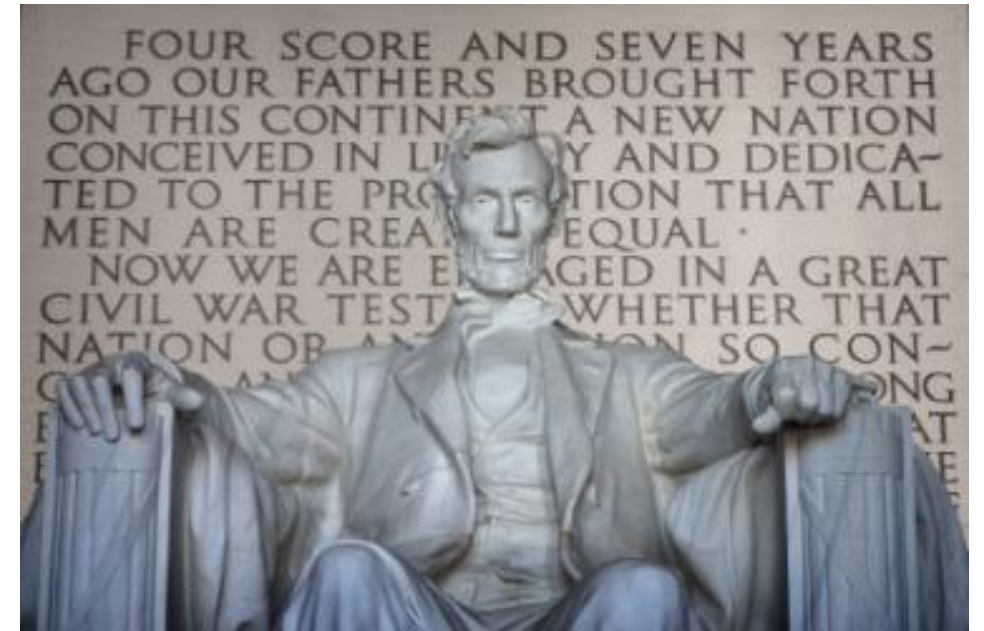
Also, keep attending the practice sessions for more practice!

Review: sampling, and sampling distributions

Review: sampling



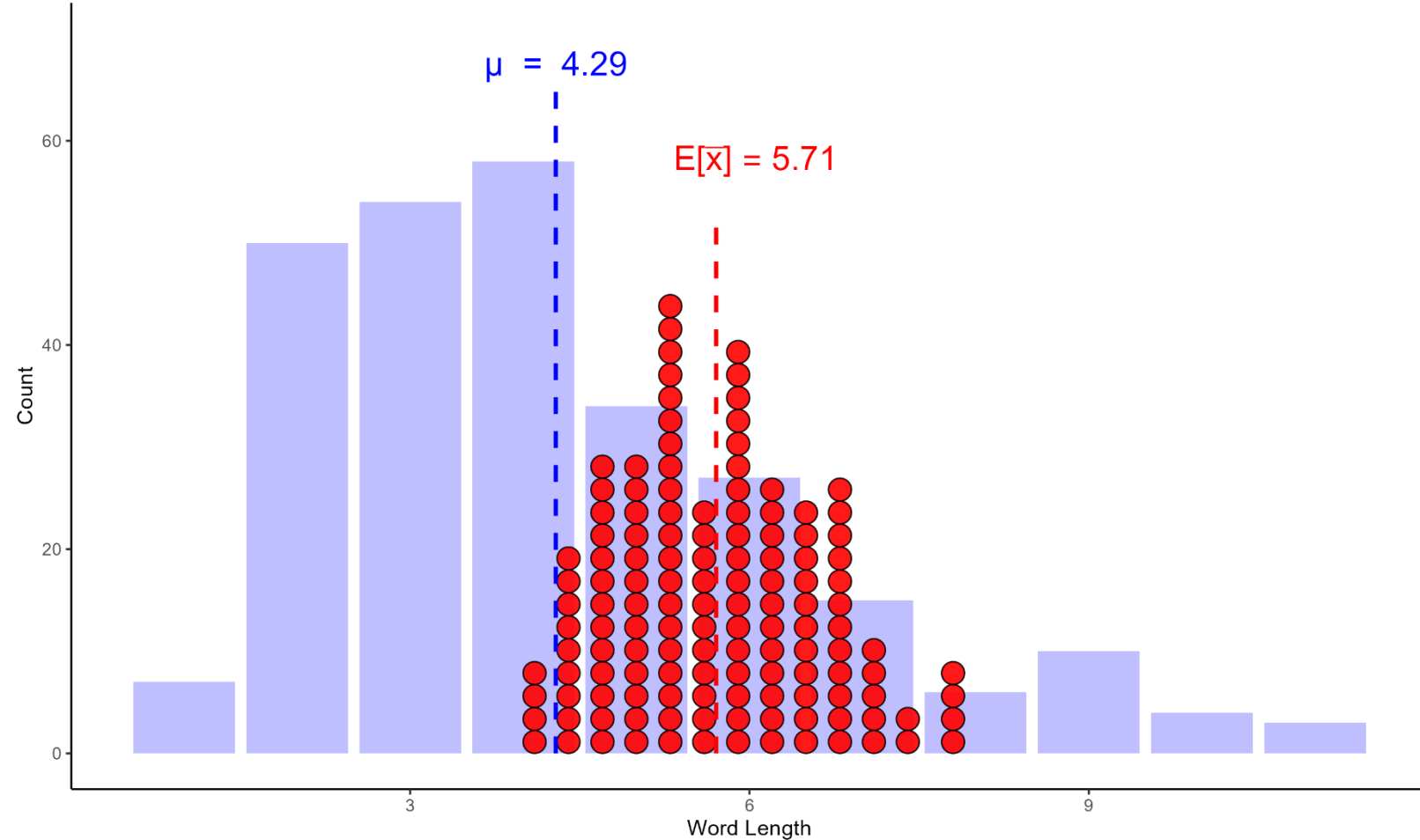
1	orange
2	red
3	green
4	white
5	white
6	white
7	white
8	white
9	red



Q: What symbol do we use to denote the sample size?

A: ***n***

Bias and the Gettysburg address word length distribution

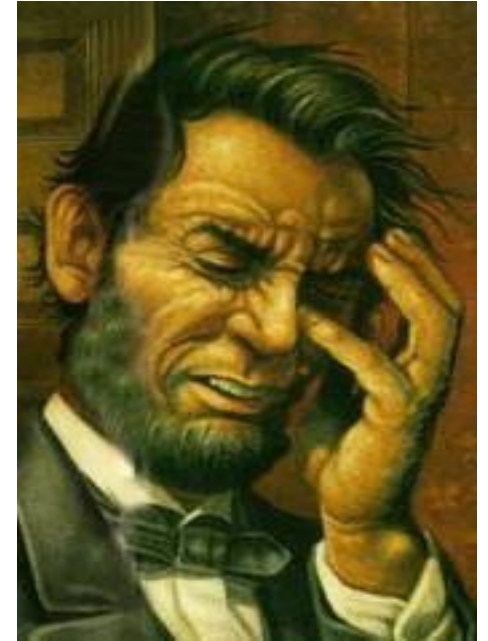
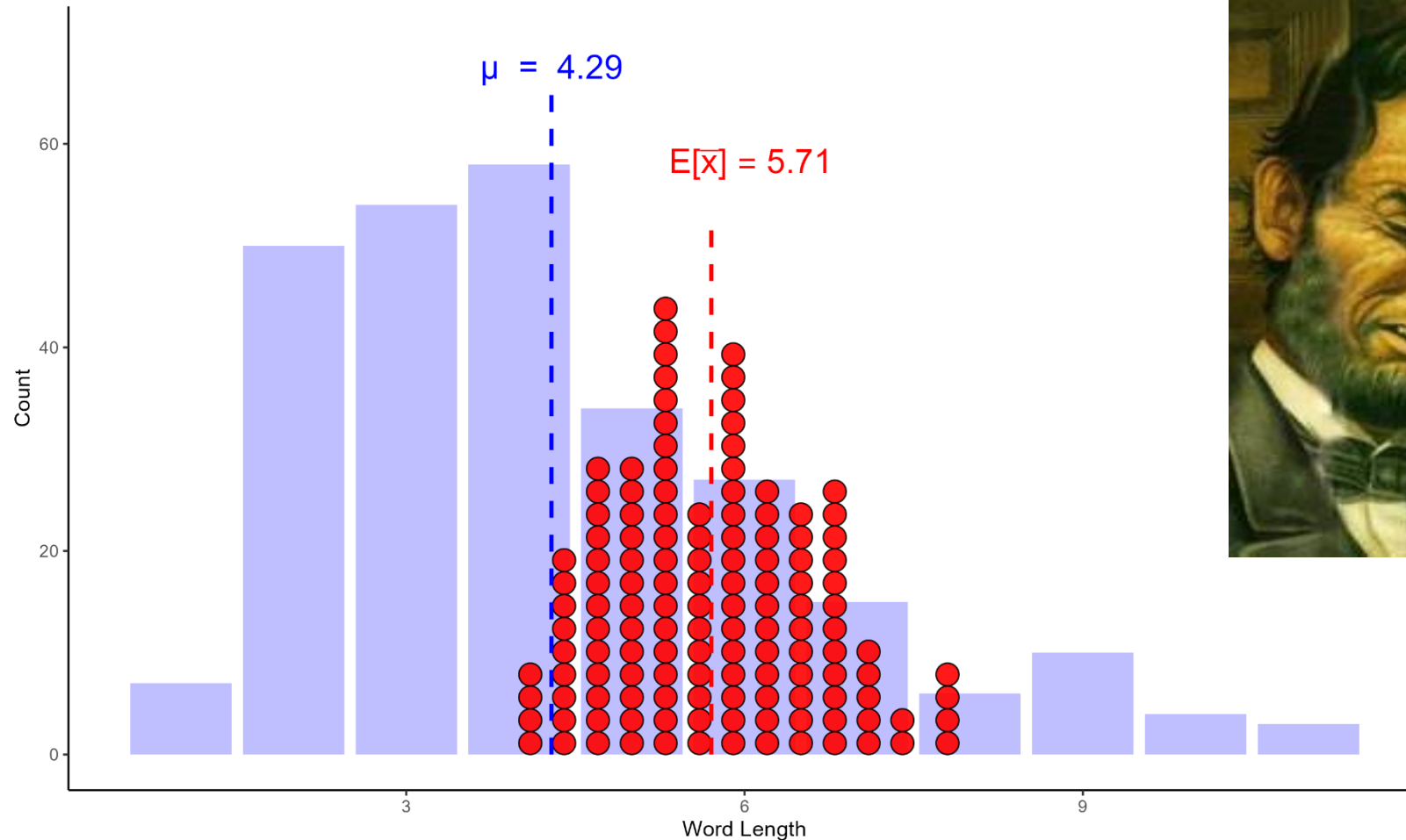


Bias and the Gettysburg address word length distribution

Bias is when the average statistic values does not equal the population parameter

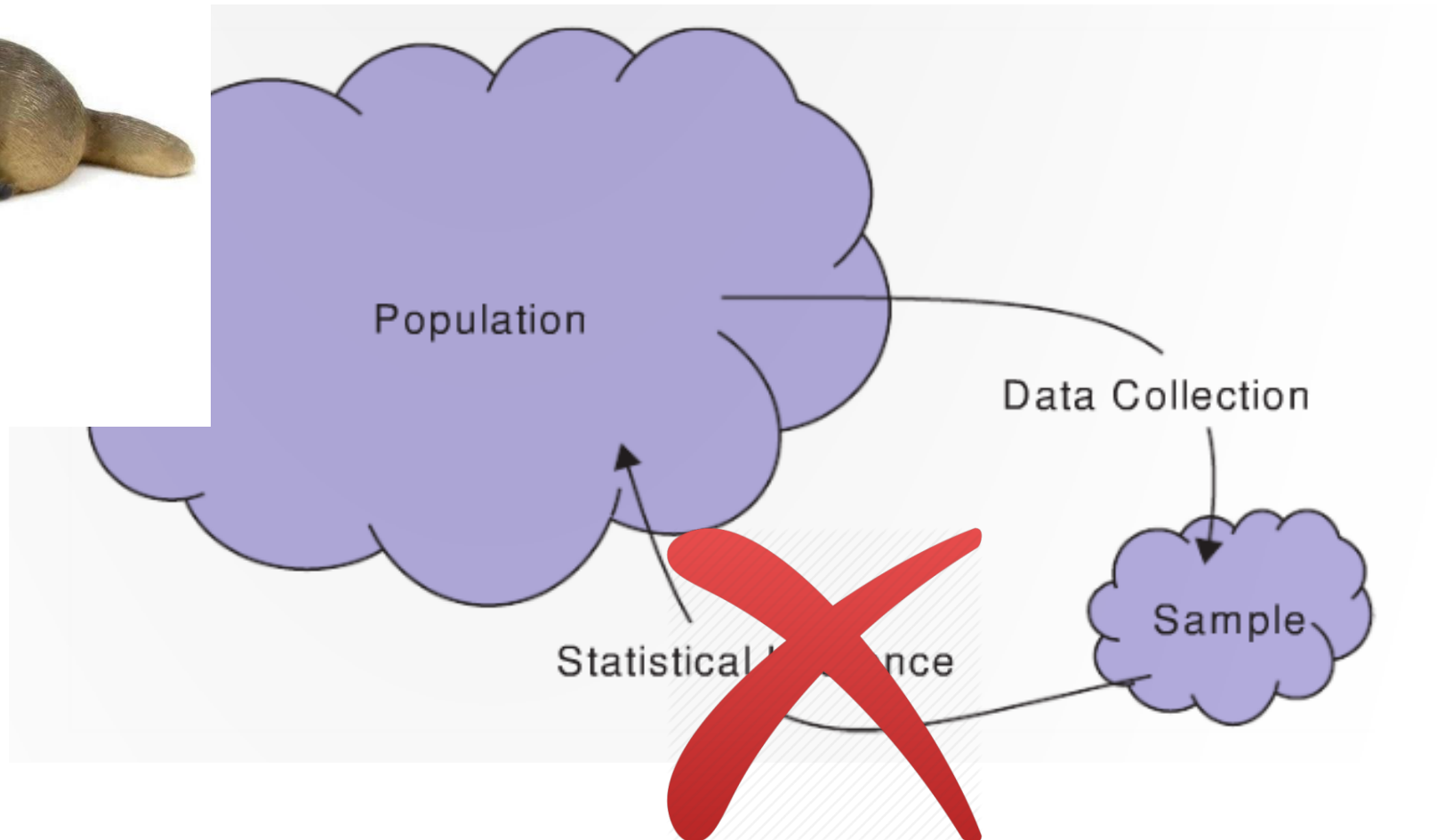
Here:

$$E_s[\bar{x}] \neq \mu$$

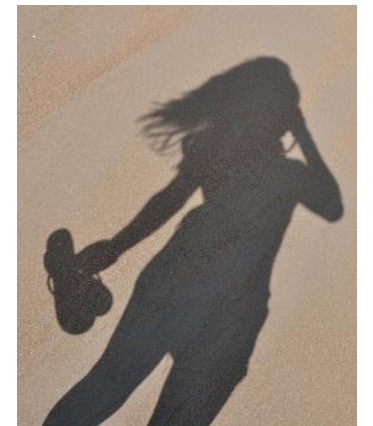


Statistical bias

μ



$\bar{x}$



Basic questions for sampling

What is the population?

What is the sample?

Do they differ in a meaningful way?





Yelp reviews of restaurants?

An anonymous survey randomly select 6,000 people and asked them if have they used an illicit drug in the past month?

<https://www.billoreilly.com/poll-center>

Also, framing the question on surveys matters

- “Majority of Americans **oppose** stricter gun control laws”
- “Majority of Americans **support** stricter gun laws”

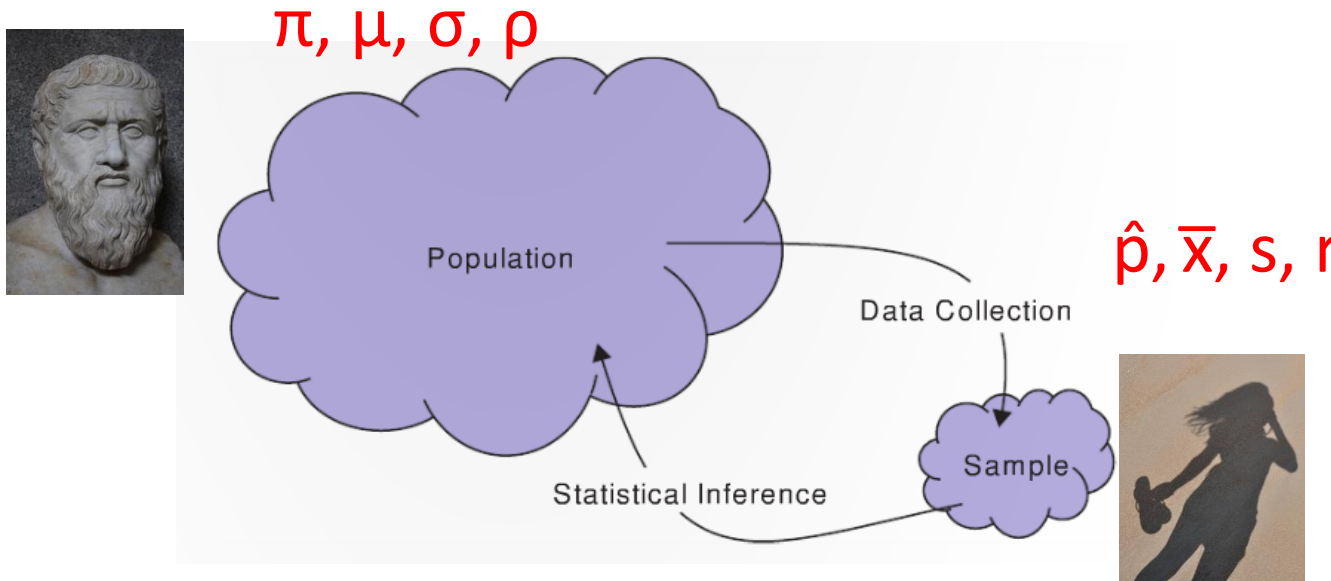
Q: How can we prevent sampling bias?

A: To prevent bias, use a **simple random sample**

- where each member in the population is equally likely to be in the sample

This allows for generalizations to the population!

Soup analogy!



Practicalities...

It might not be feasible to randomly select equally from all members of a population

This might not be a problem as long as the sample is representative of the population

Example: If we wanted to know proportion of left-handed people in the US, randomly sampling Yale students might be sufficient

Need to think carefully to avoid bias!

Statistics requires thought!

Use your own reasoning:

- What is the population I am interested in?

- Does the sample reflect the population of interest?

- Be your own worst critic!

Q: How do we select a random sample?

Mechanically:

- Flip coins

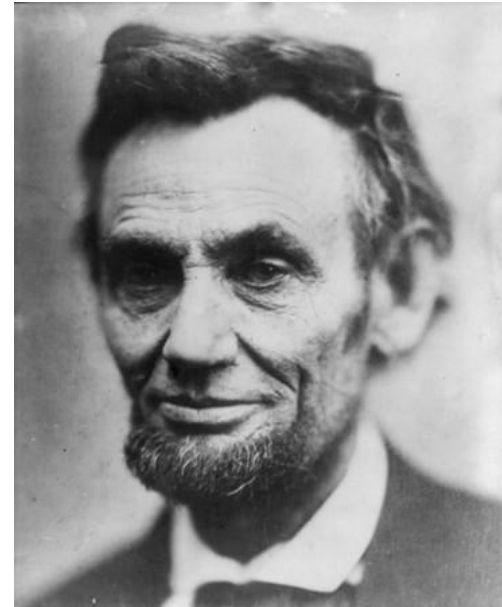
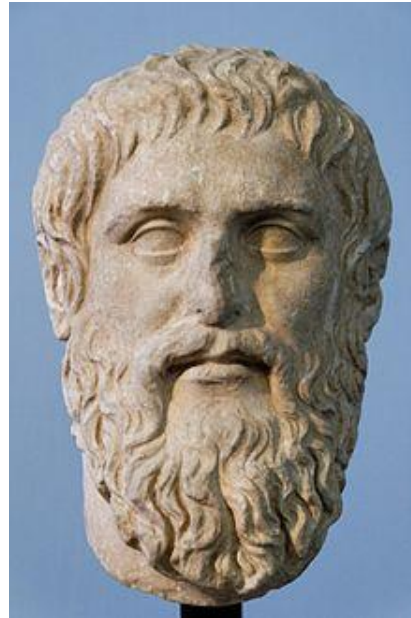
- Pull balls from well mixed bins

- Deal out shuffled cards, etc.

Use a computer program

From now on we are going to assume no bias!

Happy Plato and Lincoln

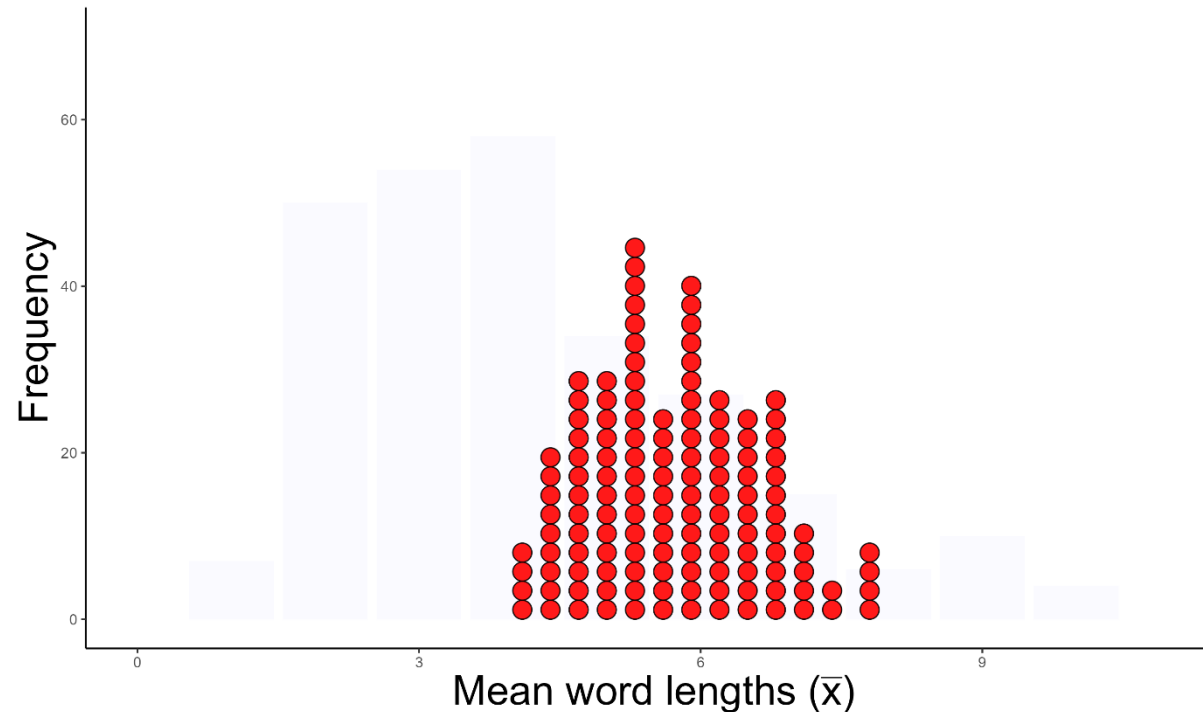


Our statistic values, on average, reflect the parameters

Review: Sampling distributions

Recall for our distribution of Gettysburg word lengths...

Q: What does each dot that is plotted correspond to?



A: The mean length of 10 words ($\bar{x}$)
i.e., each point in our distribution is a **statistic**!

Sampling distribution

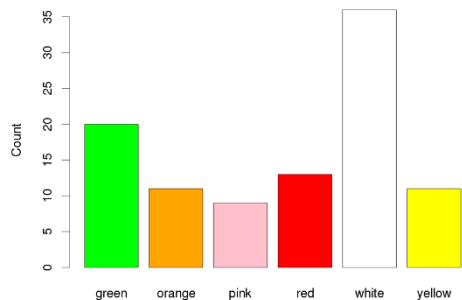
THEORETICAL CONCEPT

A **sampling distribution** is the distribution of sample statistics computed from different samples of the same size (n) from the same population

A sampling distribution shows us how the sample statistic varies from sample to sample

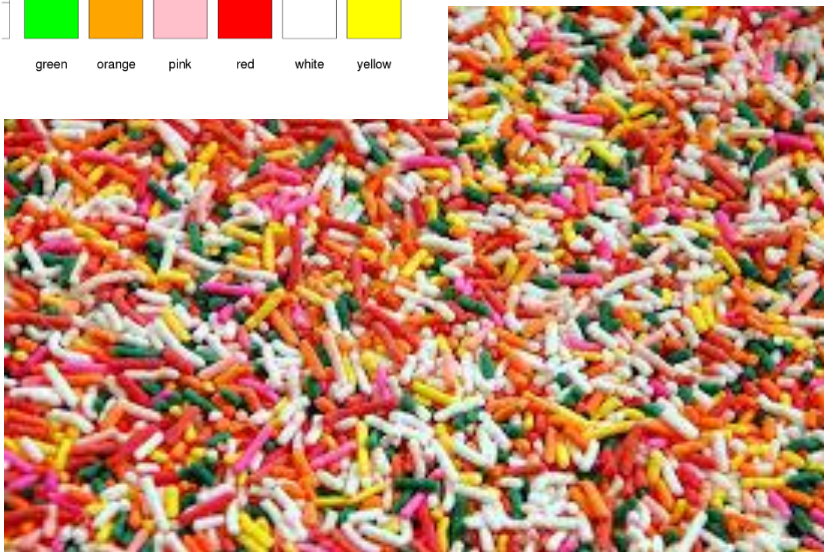
Note in general sampling distributions are a **theoretical concept** and not something that is actually created in real studies

- Although we will simulate a few sampling distributions in R so we can understand what this theoretical distribution is

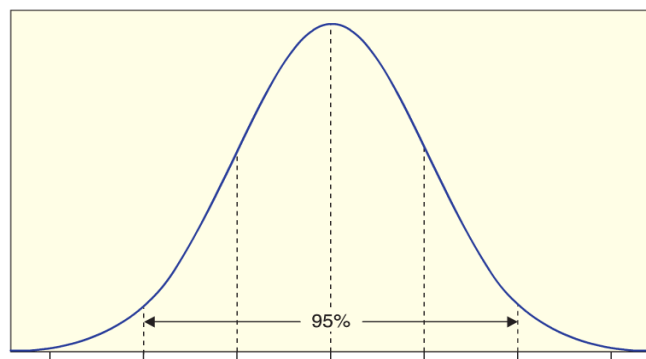


π_{red}

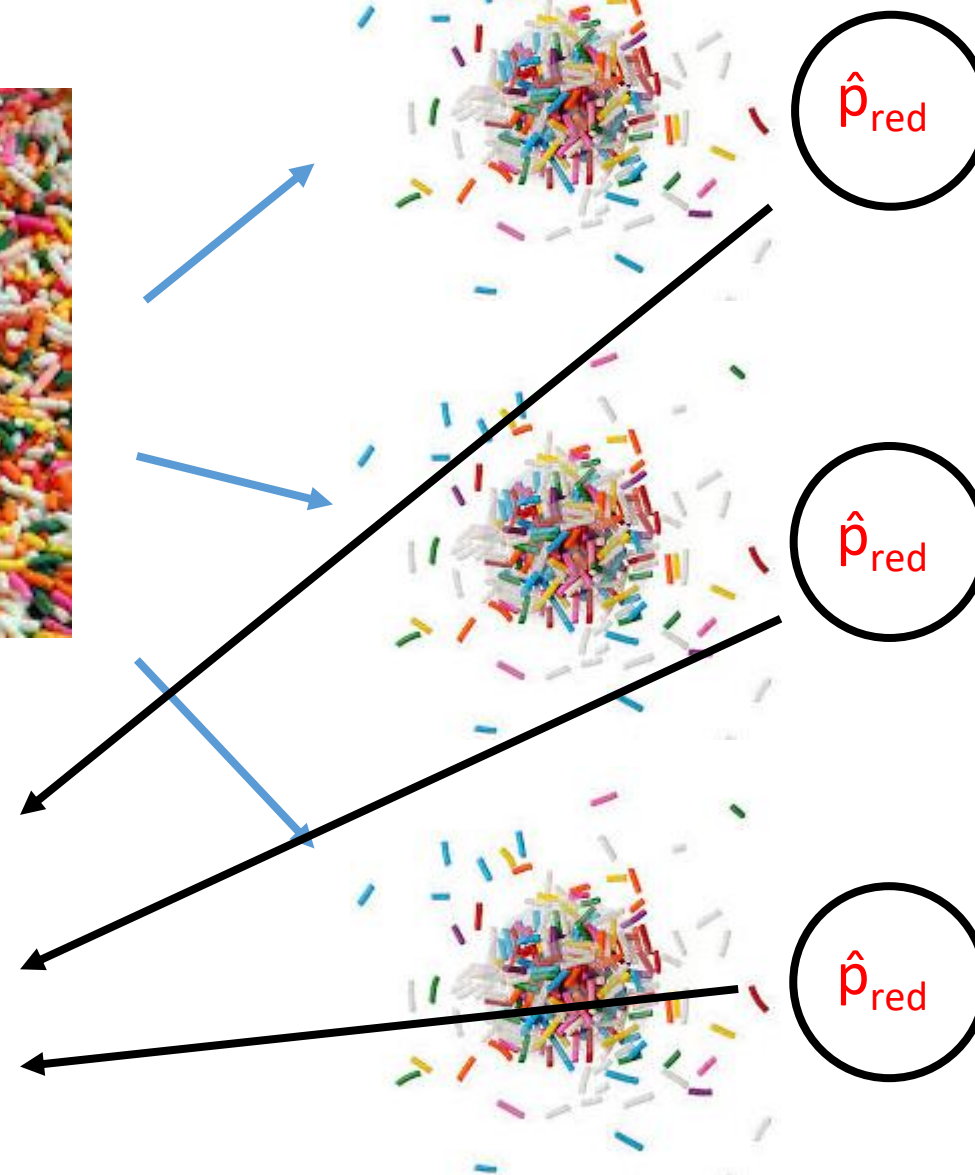
$n = 100$



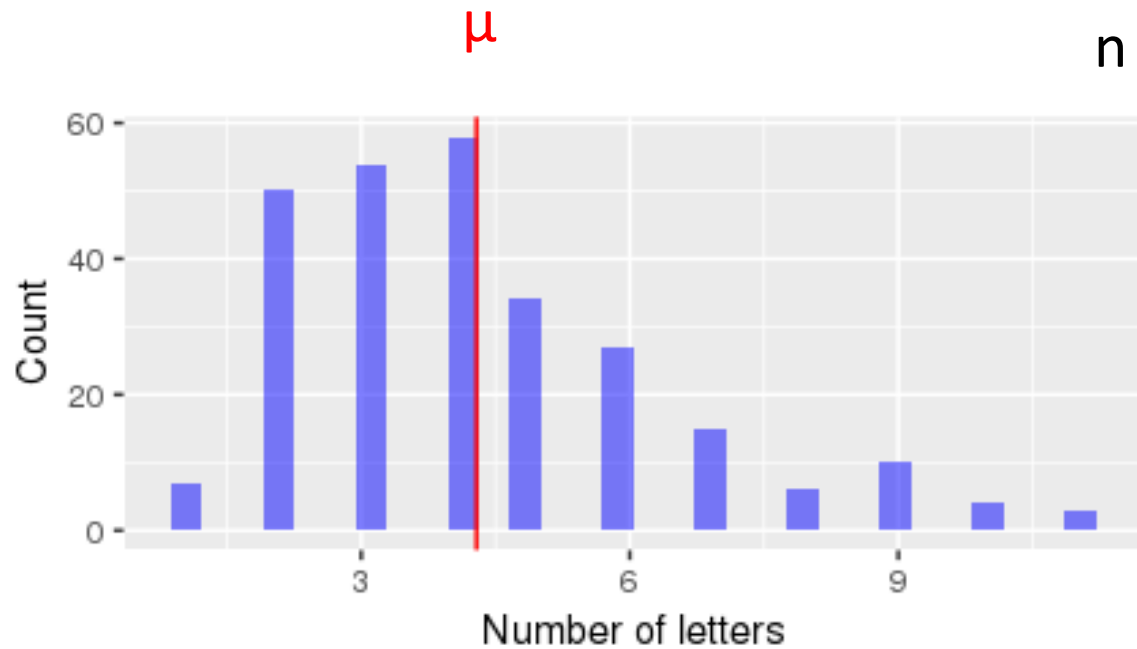
Distribution
of $\hat{p}$'s



Sampling distribution!



Gettysburg address word length sampling distribution



10, 3, 3, 3, 4,
3, 2, 6, 10, 5

$$\bar{x} = 5$$

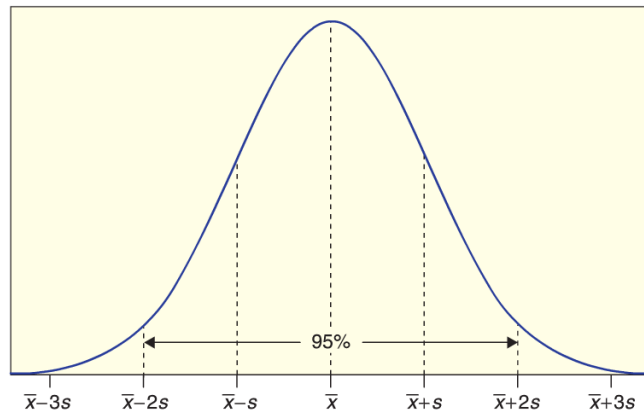
2, 6, 2, 6, 6,
2, 5, 3, 2, 9

$$\bar{x} = 4.3$$

3, 9, 3, 4, 4,
3, 6, 6, 2, 2

$$\bar{x} = 4.2$$

Distribution
of $\bar{x}$'s



Sampling distribution!

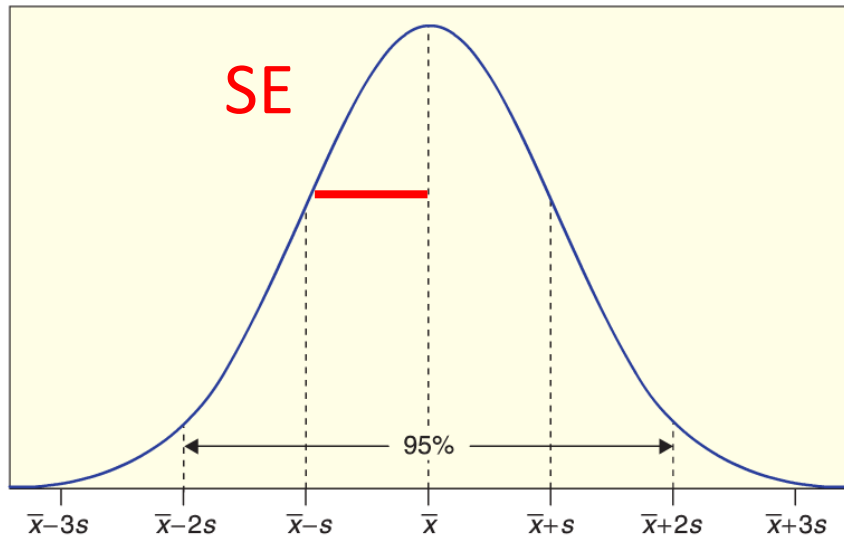
[Gettysburg sampling distribution app](#)

The standard error

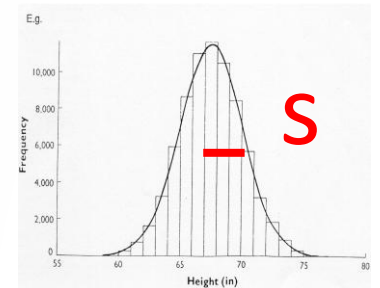
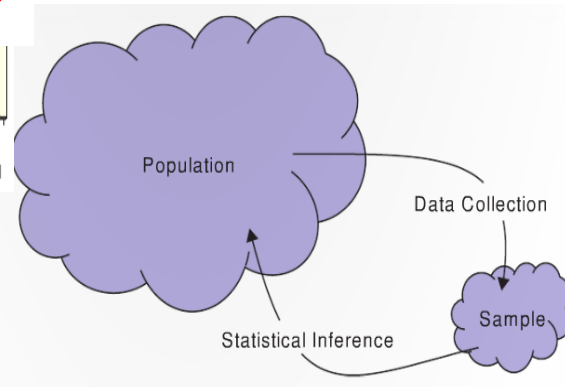
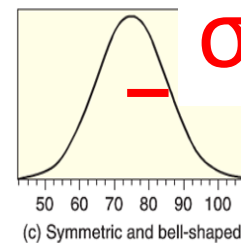
The **standard error** of a statistic, denoted SE, is the standard deviation of the sample statistic

- i.e., SE is the standard deviation of the *sampling distribution*

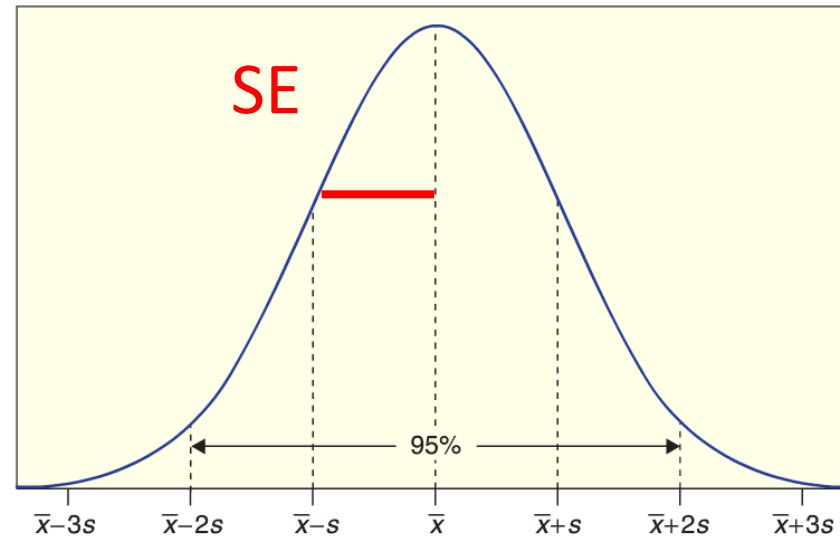
Note: standard error SE is different from σ and s !



Sampling distribution!



What does the size of a standard error tell us?



Q: If we have a large SE, would we believe a given statistic is a good estimate for the parameter?

- E.g., would we believe a particular $\bar{x}$ is a good estimate for μ ?

A: A large SE means our statistic (point estimate) could be far from the parameter

- E.g., $\bar{x}$ could be far from μ

Sampling distributions in R!

Let's create a sampling distribution in R

Load the SDS1000 library to make all SDS1000 functions available

```
library(SDS1000)
```

Get the Gettysburg population data

```
load("gettysburg.Rda")
```

```
word_lengths <- gettysburg$num_letters    # lengths of the 268 words
```

Let's create a sampling distribution in R

We can use the `sample(data_vec, n)` to get a sample of length n:

```
curr_sample <- sample(word_lengths, 10)
```

Q: How can we get $\bar{x}$ from this sample in R?

```
mean(curr_sample)
```

Q: How could we get a full sampling distribution?

- A: Repeat this many times to get an approximation of the sampling distribution
- If we store the $\bar{x}$'s in a vector, we can then plot the sampling distribution as a histogram

The `do_it()` function

The `do_it()` function (from the SDS1000 package) repeats a piece of code many times

- It returns a vector with the values created each time the code is repeated

```
do_it(100) * {
```

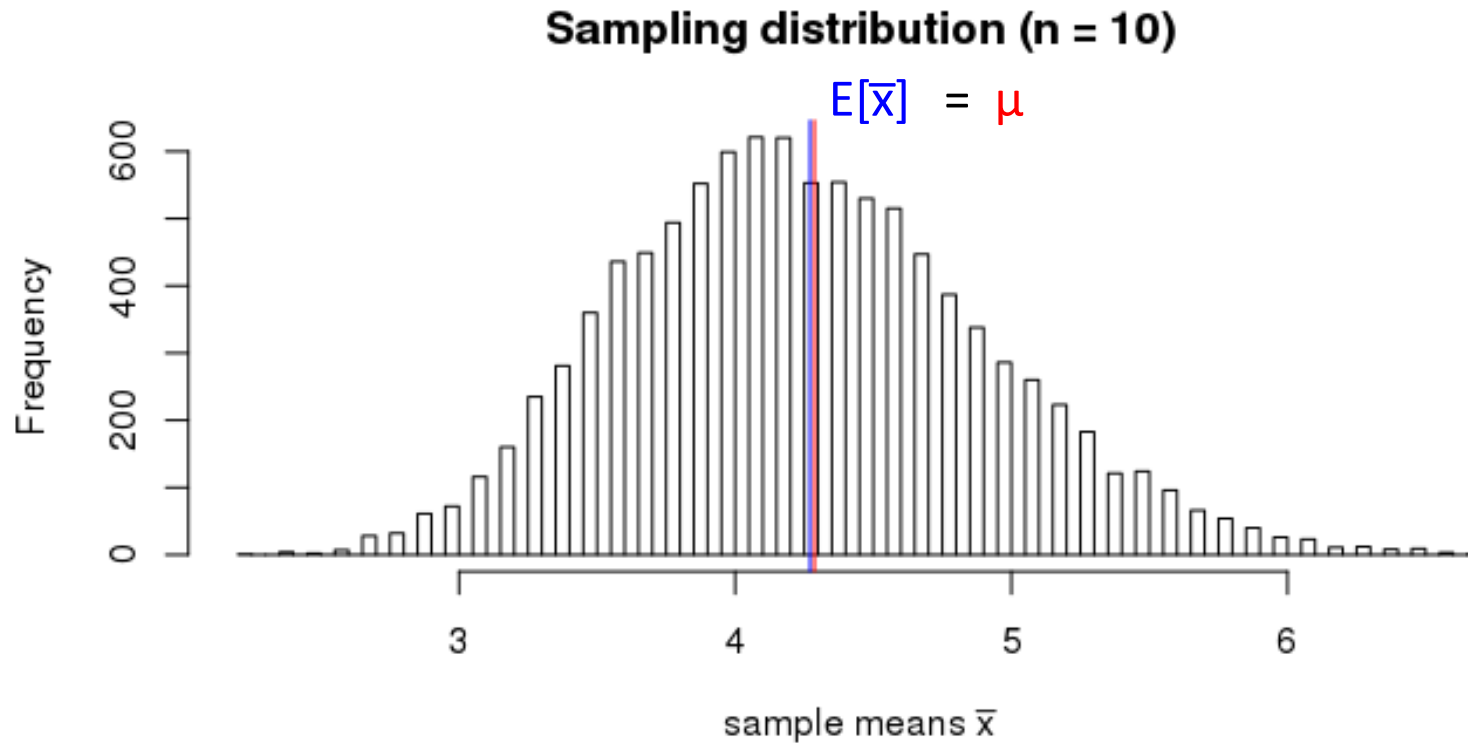
```
  2 + 3
```

```
}
```

Let's create a sampling distribution in R

```
sampling_dist <- do_it(10000) * {  
  
    curr_sample <- sample(word_lengths, 10)  
    mean(curr_sample)  
  
}  
  
hist(sampling_dist)  
SE <- sd(sampling_dist)
```

Sampling distribution in R



`mean(sampling_dist)`

`mean(word_lengths)` # these are the same, so no bias

Changing the sample size n

What happens to the sampling distribution as we change n ?

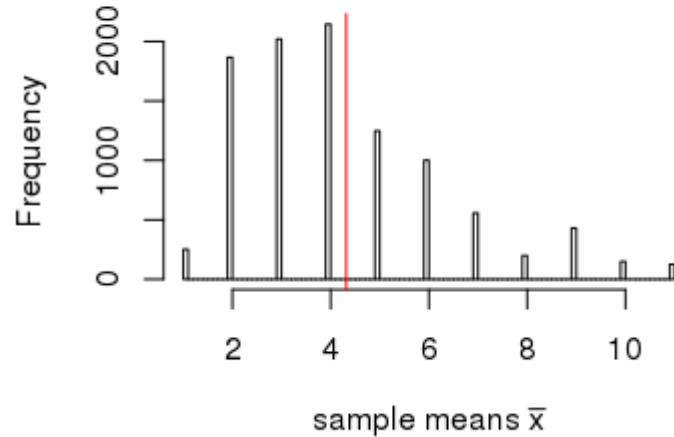
- Experiment for $n = 1, 5, 20, 80$

```
sampling_dist <- do_it(10000) * {  
    curr_sample <- sample(word_lengths, 20)  
    mean(curr_sample)  
}
```

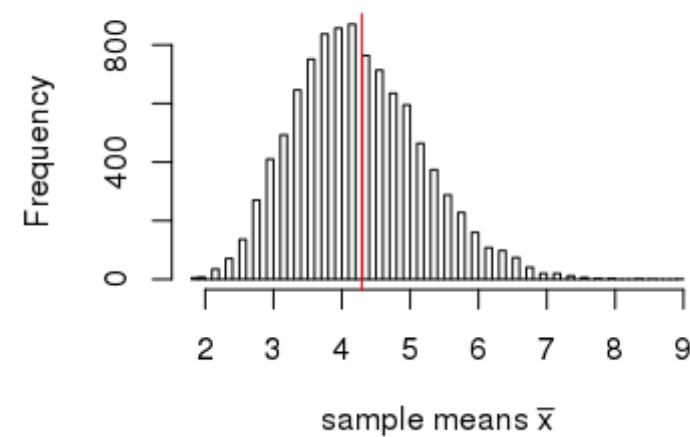
```
hist(sample_means, breaks = 100)
```

[Gettysburg sampling distribution app](#)

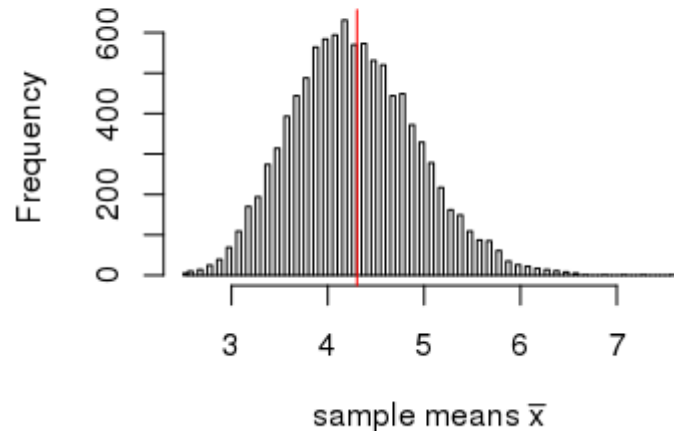
Sampling distribution ($n = 1$)



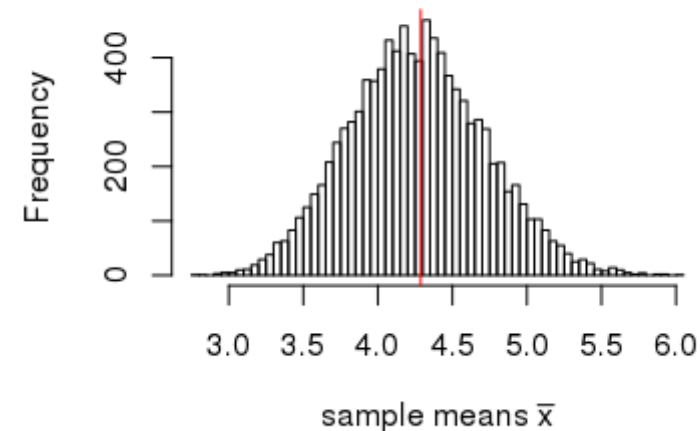
Sampling distribution ($n = 5$)



Sampling distribution ($n = 10$)



Sampling distribution ($n = 20$)



x-axis range 9 vs. 6

As the sample size n increases

1. The sampling distribution becomes more like a normal distribution
2. The sampling distribution points ($\bar{x}$'s) become more concentrated around the mean $E[\bar{x}] = \mu$

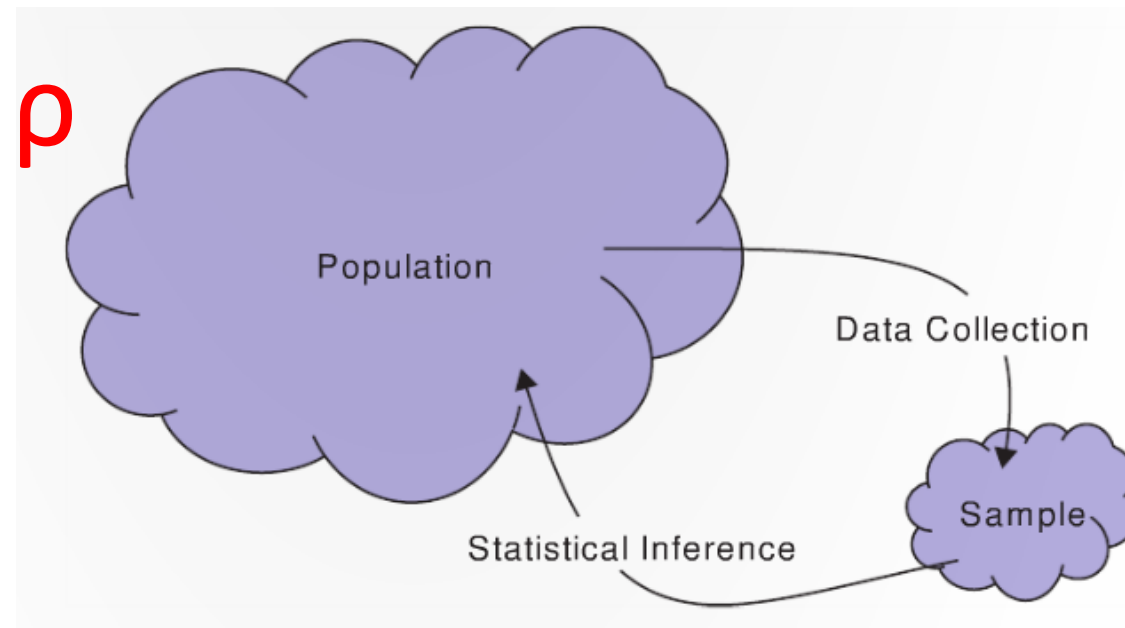
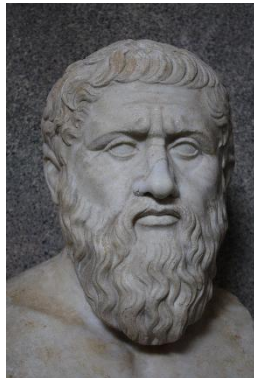
Point estimates and confidence intervals

Back to the big picture: Inference

Statistical inference is...?

the process of drawing conclusions about the entire population based on information in a sample

π, μ, σ, ρ



$\hat{p}, \bar{x}, s, r$



Point Estimate

We use a statistic from a sample as a **point estimate** for a population parameter

- $\bar{x}$ is a point estimate for...?

40% of Americans approve of Trump's job performance according to a poll of 2,464 people

Q: What are π and $\hat{p}$ here?

Q: Is $\hat{p}$ a good estimate for π in this case?

Interval estimate based on a margin of error

An **interval estimate** give a range of plausible values for a population parameter

One common form of an interval estimate is:

$$\textit{Point estimate} \pm \textit{margin of error}$$

Where the **margin of error** is a number that reflects the precision of the sample statistic as a point estimate for this parameter

Example: Gallup poll

40% of American approve of Trump's job performance, plus or minus 3%

How do we interpret this?

Says that the population parameter (π) lies somewhere between 37% to 43%

i.e., if they sampled all voters, the true population proportion (π) would likely be in this range

Confidence Intervals

A **confidence interval** is an interval computed by a method that will contain the *parameter* a specified percent of times

- i.e., if the estimation were repeated many times, the interval will have the parameter x% of the time

The **confidence level** is the percent of all intervals that contain the parameter

Think ring toss...

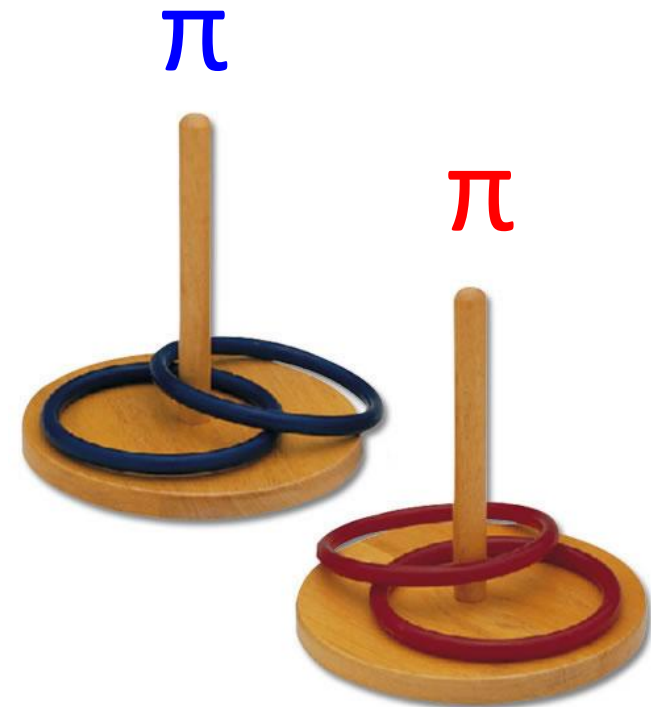
A parameter exists in the world

We toss intervals at it

- These are our confidence intervals

95% of those intervals capture the parameter

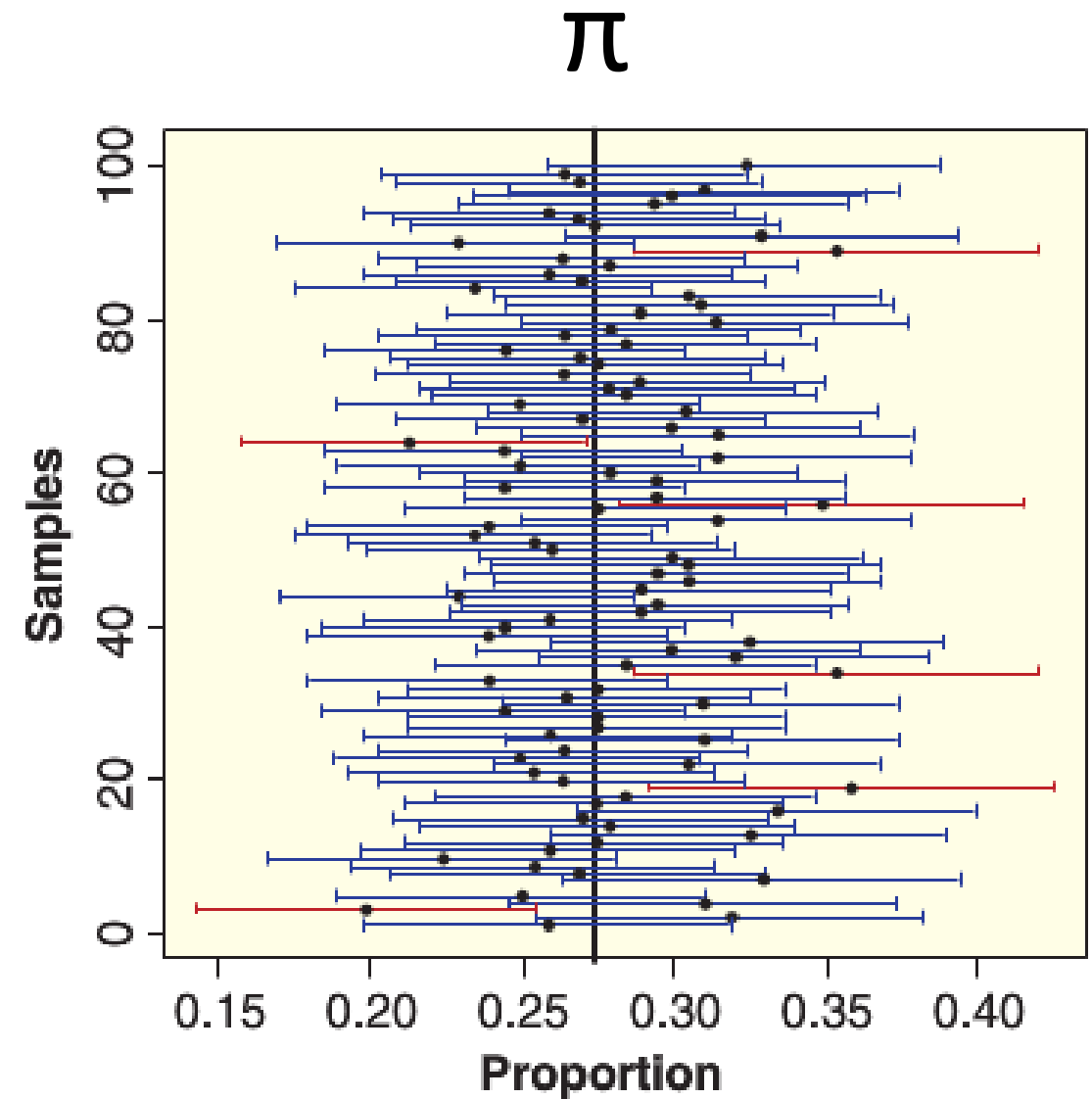
- (95% is our confidence level)



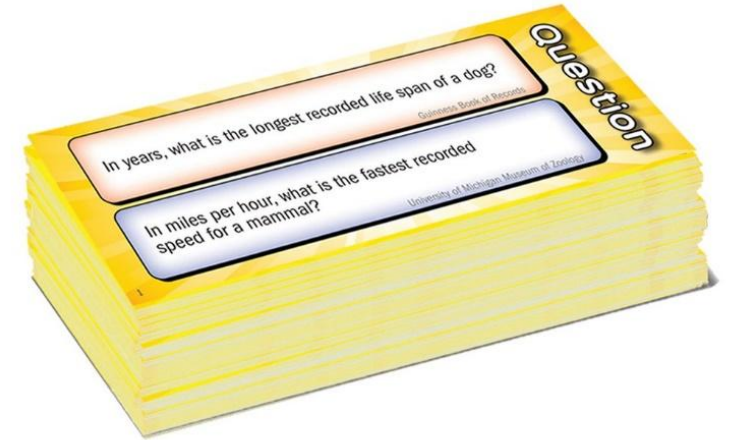
Confidence Intervals

For a **confidence level** of 95%...

95% of the **confidence intervals** will have the parameter in them



Wits and Wagers: 90% confidence interval estimator



I will ask 10 questions that have numeric answers

Please come up with a range of values that contains the true value for 9 out of the 10 questions

- I.e., be a 90% confidence interval estimator

Write your range of estimates for each question as two numbers

- E.g., [10.2 to 50.7]